

Retrieval Grounding Matters Most Where Model Memory Is Thinnest: An Agentic Retrieval Augmented Advisory System Evaluated Across Two Filing Markets

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Abstract. Context. Retrieval augmented generation gets evaluated on United States corporate filings almost without exception. An assumption rides along unexamined: that the model underneath knows nothing of the documents it is handed.

Approach. CyberBoard-AI was run against two markets rather than one. BM25 sparse scoring fuses with dense embeddings. A cross encoder reranks. A ReAct agent holding three retrieval tools sits above the pipeline. Both baselines are ablations of the same codebase, and every configuration meets identical questions.

Corpora. 116 SEC EDGAR filings from 20 companies, cut into 18,136 chunks. Against that, 845 chunks from seven Bahrain Bourse issuers, carrying 35 hand verified questions.

Findings. Without retrieval, the closed book model answers 63.2 percent of recent United States questions and 11.4 percent of Gulf Cooperation Council questions. The agent reaches 96.8 percent accuracy and 99.4 percent faithfulness over five runs, against 82.9 percent for single pass retrieval augmented generation. McNemar returns five wins and zero losses, at an exact p value of 0.0625. Reranking gains 12.9 percentage points on the larger corpus and loses 2.9 on the smaller.

Implications. A baseline that answers from memory is no floor, so single market evaluation understates what retrieval contributes. Reranking belongs in per corpus configuration rather than in a default. And a faithfulness judge left unexamined misreported the metric it existed to measure, until a context truncation defect surfaced.

Availability. Code, data settings, and every result file are released at a pinned commit.

Keywords: retrieval augmented generation, agentic systems, financial question answering, cross market evaluation, hybrid retrieval, explainability.

1 Introduction

A board overseeing a portfolio reads filings for a living. One annual submission to the United States Securities and Exchange Commission can run to several hundred pages,

and the obligation to act on what it contains is a regulatory one. The Central Bank of Bahrain Corporate Governance Code [29] frames board responsibility around informed oversight, which presumes directors can reach the information their oversight depends on.

Retrieval augmented generation [1] offers an obvious route to that information, and the financial variant of the problem has attracted careful work [3], [4]. Almost all of it is built and measured on United States disclosure. Whether the resulting designs transfer to a smaller regional market is not a settled question, and for an institution obliged to read both there is little published evidence to consult.

This paper reports what happened when one system was evaluated on both. The finding that motivates the title is not about the system at all. It concerns the baseline. A closed book model with no retrieval answers 63.2 percent of questions on recent United States filings and 11.4 percent of matched questions on Bahrain Bourse annual reports. Disclosure from very large United States issuers is heavily represented in the text such models are trained on; Bahrain Bourse reports are not.

Two consequences follow. Any evaluation confined to United States data understates what retrieval contributes, because the model can partly answer without it and the baseline therefore looks stronger than it is. And where model memory is thin for a market, retrieval grounding is not a refinement of the capability but the whole of it, so the argument for building such systems is strongest precisely where training data is weakest. Neither claim is available from a single corpus.

Four contributions follow from that. The first is the cross market evaluation itself: one agentic retrieval augmented system measured on a United States benchmark and on hand verified Gulf Cooperation Council filings, with the closed book gap treated as a finding rather than as a baseline row.

Reranking supplies the second. Cross encoder reranking turns out to be corpus dependent rather than uniformly beneficial, gaining 12.9 percentage points on the larger corpus and losing 2.9 on the smaller and more homogeneous one.

Third comes a retrieval defect, documented here together with its correction, which moved the FinanceBench hit rate from 54.8 percent to 96.8 percent. It is offered as a transferable lesson about testing end to end behaviour rather than units alone.

The last is a caution about measurement. An unexamined faithfulness judge misreported faithfulness on the corpus whose evidence ran longest, and inspection rather than the evaluation protocol is what caught it.

2 Related Work

2.1 Retrieval Augmented Generation over Financial Text

Lewis et al. [1] pair parametric knowledge with non parametric retrieval, and Gao et al. [2] later sort the field into naive, advanced, and modular paradigms. Their criticism of the naive form is the relevant part here: retrieve once, concatenate, and there is no route to recovery when the first retrieval goes badly.

Within finance, FinSage [3] builds a two phase multi path search with reranking and reports high recall. Lithgow-Serrano et al. [4] survey the area more broadly and locate the variance in chunking and matching rather than in the choice of generator. That finding shaped the present design, and it proved accurate: a defect in retrieval dominated everything downstream of it. Outside finance, case based legal retrieval [5] and a Vietnamese legal study [6] both establish sparse plus dense hybrid retrieval with reranking in document rich regulated domains structurally close to corporate governance.

2.2 Agentic Architectures

ReAct [7] interleaves reasoning traces with actions and leaves the reasoning inspectable. Toolformer [8] shows a model can learn when to call an external tool. Surveys by Wang et al. [9] and Hu et al. [10] place this style of tool calling at one point in a design space extending to multi agent routing, exemplified by Tool to Agent Retrieval [11] and MasRouter [12].

The surveys also register a gap this paper takes seriously. Agentic work reports capability far more often than controlled comparison. Demonstrations that an agent can complete a task are common; measurements of how much the agentic layer contributes over a matched retrieval baseline, on identical questions scored by the same judge, are not. That comparison is the central experiment reported in Section 5.

2.3 Retrieval Infrastructure

BM25 rests on the probabilistic relevance framework of Robertson and Zaragoza [13]. Exact term matching is where it is strong. Paraphrase is where it is blind: a query that says the same thing in different words gets no credit for it.

Dense retrieval carries the mirror image of that profile, and Sentence-BERT [14] is where the relevant design originates. Its motivation is worth restating because it shapes the architecture used here. A cross encoder reads a query and a passage jointly, which is what makes it accurate, and that same joint reading is what makes it unusable across a corpus, since every candidate demands a full model pass. Reimers and Gurevych answered by training a bi encoder that maps each passage to a vector once, in advance, leaving retrieval as a cheap similarity computation.

Neither model type escapes that tension alone, which is why the system uses both in sequence. The bi encoder, paired with the HNSW index of Malkov and Yashunin [15], sweeps the corpus cheaply and returns a shortlist. The cross encoder then reads that shortlist properly and reorders it, so expensive accuracy is spent only where it can change the outcome. Pradeep et al. [16] set out technique for the reranking stage and are equally clear that it is not free: a reranker can demote a correct passage. Section 5.3 reports that happening.

2.4 Evaluation and Faithfulness

FinQA [18], TAT-QA [19] and ConvFinQA [20] define what the field treats as hard in numerical reasoning over filings, and PACIFIC [21] adds proactive clarification. FinanceBench [17] is used here instead, because the first question this work asks is a retrieval question and FinanceBench pairs questions with the evidence passages that answer them. On the measurement side, Zheng et al. [22] establish the LLM as judge methodology, Es et al. [24] decompose evaluation without references, Liu et al. [23] treat judge consistency as a property to be measured, and Song et al. [25] and Sriraman et al. [26] approach hallucination from tuning and from internal signals respectively. Read together they describe an instrument still under construction, which is why the judge in this study was validated rather than trusted.

3 System Design

The system is a retrieval pipeline with an agent above it, and the split between offline and online work carries most of the design.

Offline, the pipeline downloads each filing, parses it into named regulatory sections, chunks every section recursively at 512 tokens with 64 token overlap, embeds the chunks and indexes them. None of that happens while a reader waits. The chunk size is a compromise in both directions: cut too small and a figure is stranded from the row heading that identifies it, which is fatal in financial disclosure, while cutting too large dilutes the embedding, since one vector asked to carry several distinct claims carries none of them sharply.

At query time the retrieval layer fuses BM25 sparse scores with dense similarity at 0.7 dense to 0.3 sparse, both normalised to a zero to one range first so that two differently scaled signals can be added at all. The weighting favours semantic matching because most questions arrive in natural language rather than in the vocabulary of the filing. The sparse component keeps a meaningful share regardless, since tickers, line item names and defined terms are exactly where exact matching beats embedding similarity. A cross encoder then reranks the fused candidates.

Three function calling tools sit above the pipeline, and a ReAct agent selects among them from the question rather than routing every query through one general retrieval call.

A financial summarizer handles the quantitative side, covering revenue, earnings and comparable filing figures. A risk identifier pulls disclosed risk factors and narrows them by category, whether regulatory, market, operational, cybersecurity or supply chain. A compliance checker surfaces governance, internal control and audit disclosures. All three share the one hybrid retrieval pipeline; what differs is that each scopes its query to the company in question.

Whatever the agent returns arrives wrapped the same way every time: a source attribution, a per source confidence, and an advisory disclaimer. Where a tool returns no passages the agent is left with nothing to ground an answer in, and it says so rather than reaching into parametric memory. That abstention is implemented rather than

merely requested, and it is what makes the comparison against the closed book model a comparison of behaviours and not only of accuracies.

Two design details carry more weight than their size suggests. Chunk identifiers encode ticker, filing type, filing date, section, and an index, so scoping a query to one company reduces to a prefix test rather than a metadata lookup. The two corpora sit in separate collections rather than in one store with a market field, which makes cross market leakage structurally impossible rather than a matter of correct filtering. The cross market claim in Section 5 depends on the two being measured independently.

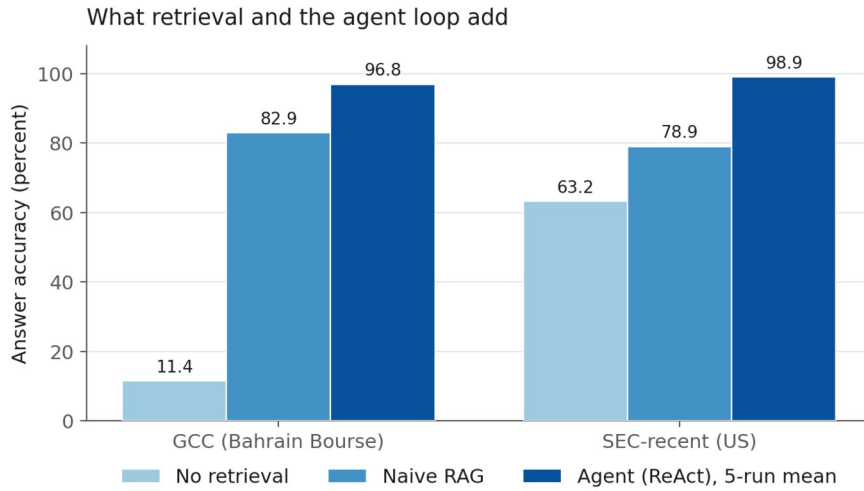


Fig. 1. Accuracy by configuration across both corpora. Agent values average five runs; baseline values come from one run over identical items.

4 Experimental Setup

The United States corpus is 116 SEC EDGAR filings of type 10-K and 10-Q from 20 companies, cut into 18,136 chunks. The Gulf Cooperation Council corpus is a set of Bahrain Bourse annual reports, 845 chunks, drawn from seven issuers: National Bank of Bahrain, BBK, Aluminium Bahrain, Beyon, GFH Financial Group, Bahrain Islamic Bank, and Kuwait Finance House Bahrain. Banking dominates that exchange, so the set is weighted toward it, with an industrial issuer and a telecommunications group included so the questions are not purely about banking.

Retrieval quality is measured as hit rate against the FinanceBench matched questions, a retrieved set counting as a hit when it holds the gold evidence. The bluntness of that metric is deliberate rather than a concession. Ranking measures such as mean reciprocal rank reward a system for placing the right passage first rather than third, which matters when a human scans a result list. That is not the situation here. The retrieved set goes to an agent that reads all of it, so what decides the outcome is whether the evidence is present, not where in the ordering it sits.

End to end answer quality is measured on 35 hand verified Gulf Cooperation Council questions, five for each issuer, with a further 19 numerical questions on recent United States filings serving as an independent check. Each gold answer was traced back to the statement it came from before being trusted as ground truth. Three proved wrong during that pass and were corrected, which is worth reporting because it establishes that the instrument is fallible.

Correctness and faithfulness are scored separately by an LLM judge at temperature zero, in the manner of Zheng et al. [22]. Separating them is what makes the error analysis in Section 5.4 possible, because an answer can be faithful to its evidence and still wrong. Two baselines are ablations of the same codebase rather than separate implementations: a closed book model with no retrieval, and a single pass retrieval augmented model that retrieves once and answers without agent or tool selection.

5 Results

5.1 Cross Market Baseline Gap

The headline comparison is between the two baselines rather than between the system and either of them. On recent United States questions the closed book model reaches 63.2 percent accuracy. On Gulf Cooperation Council questions it reaches 11.4 percent. The system is identical in both cases and so is the prompt; what differs is whether the answers are already present in the model.

That fivefold gap is the paper's main empirical claim. It says that a retrieval evaluation confined to United States filings measures a weaker version of the problem, because the closed book condition is not a genuine floor there. It also says that the case for retrieval grounding is strongest in markets that published benchmarks currently ignore.

5.2 End to End Answer Quality

Averaged over five runs on the Gulf Cooperation Council set, the full agent reaches 96.8 percent accuracy with a standard deviation of 0.1, and 99.4 percent faithfulness with a standard deviation of 1.2. Single pass retrieval reaches 82.9 percent on the same items.

Five runs rather than one are reported because the boundary between local and remote computation is also the boundary where exact reproducibility ends. Retrieval is deterministic: the same query against the same index returns the same candidates in the same order. Generation is served by a third party API and offers no such guarantee. That asymmetry is why retrieval figures hold steady to a tenth of a point while answer figures carry a visible standard deviation.

The same agent was run over the 19 question United States numerical set as an independent check that the result is not an artefact of one corpus. It returned 98.9 percent accuracy and 96.8 percent faithfulness, which indicates the answer quality result transfers across the two markets.

Table 1. Answer quality over the 35 item Gulf Cooperation Council set. Agent values average five runs; baseline values come from one run over identical items.

Configuration	Accuracy	Faithfulness	Hallucination	Mean latency
No retrieval (closed book)	11.4%	0.0%	100.0%	6.54 s
Single pass retrieval	82.9%	97.1%	2.9%	6.56 s
Full agent (proposed)	96.8%	99.4%	0.6%	9.25 s

The closed book faithfulness of zero is definitional rather than surprising. With no retrieved context there is nothing for an answer to be grounded in, so every closed book answer is logged ungrounded.

5.3 Reranking Is Corpus Dependent

On the United States corpus, hybrid retrieval with reranking reaches 96.8 percent and without it 83.9 percent, so the reranking stage contributes 12.9 percentage points. On the Gulf Cooperation Council corpus the ordering reverses: without reranking the hit rate is 100 percent and with it 97.1 percent.

Size explains part of that reversal and homogeneity explains more of it.

The United States corpus holds 18,136 chunks drawn from twenty companies across several industries, so a query has many plausible near matches and the cross encoder earns its cost by discriminating among them. The Gulf Cooperation Council corpus holds 845 chunks from seven issuers, most of them banks filing structurally similar statements. There the correct passage is frequently at the top of the fused ranking already, and a reranker asked to improve on an ordering that is already correct has nothing to win and one position to lose.

Read that way the finding is not a defect in the reranker but a boundary condition on it. Cross encoder reranking pays when the candidate pool is large and genuinely ambiguous, and stops paying when the pool is small and uniform. A system serving both markets should therefore treat reranking as a per corpus setting rather than a global default, which is a concrete design consequence this evaluation produced and the literature anticipated only in the abstract.

5.4 Error Analysis

How the single pass baseline fails is easily misdiagnosed, and the misdiagnosis matters because it points remediation in the wrong direction. Six of its thirty five answers are wrong, giving 82.9 percent accuracy. Faithfulness over the same set stands at 97.1 percent, with a single answer logged as a hallucination.

Those two numbers do not sit together comfortably at first glance. A system getting one answer in six wrong might be expected to be inventing things. It is not. Grounding

worked, and worked almost perfectly: the baseline says what its retrieved passages support and little else. What it cannot do is find the passage that matters. Every one of its failures is a search failure, and a fabrication check will never surface them because nothing was fabricated.

The five questions on which the agent succeeded and the baseline failed fall into three kinds, and setting them out individually is more useful than reporting the count.

Two are honest retrieval misses. In both, the baseline returned an explicit statement that the figure was not available in the retrieved context. That is the correct behaviour given what it was handed, and it is a failure of search rather than of reasoning.

Two more are multi part questions answered halfway. One asked for a chief executive and a board chair; the chief executive came back correct and the chair came back wrong. The other asked for owner equity and earnings per share; the equity figure was right and the earnings per share was reported unavailable. A question answered in half is still a question answered wrongly, and a single retrieval has no second attempt to spend on the second quantity.

Case five is the one worth dwelling on, because it defeats the checks that exist. The question sought a total income figure. The baseline supplied a real number, drawn from the issuer's own report, and attributed it accurately to the page it came from.

It was still wrong. The number came from the five year financial summary rather than from the audited statement of income. Two tables, one document, both entirely plausible, and only one of them answering the question that was asked. No hallucination detector catches this, because nothing was invented; no faithfulness judge catches it either, because the answer is faithful to the passage retrieved. The error is upstream of both.

Taken together the five cases describe one problem rather than five. Each is a retrieval failure wearing the costume of a reasoning failure, and a scoped tool call is what addresses it: the agent can go after the specific quantity a question turns on, and go again for a second quantity when a question asks for two. The five also fall across five separate issuers, so no single report or filing style explains the pattern.

5.5 Statistical Significance

Both systems answer identical items, so the comparison is paired and a McNemar test is the appropriate instrument. Anything unpaired would discard the pairing that makes the comparison informative.

On the primary run, 33 of the questions were scored by both systems. The two agreed on most and disagreed on five. Every one of those five disagreements ran the same way, with the agent correct and the baseline wrong, and none ran the other way. The exact two sided p value on that configuration is 0.0625. Conventional practice wants 0.05, so the result misses.

What the miss measures is sample size rather than effect size, and the arithmetic makes that plain. Where every discordant pair falls in one direction, the exact two sided value reduces to twice one half raised to the number of pairs. Five pairs give 0.0625. Six would give 0.03125.

One further question resolved in the agent's favour would therefore have carried this result across the conventional threshold, with nothing whatever changed about the system being tested. A test that cannot reach significance on five pairs is still describing a clean sweep. It is the evaluation set that is too small to certify an effect whose direction is not in doubt, which is why Section 8 places a larger annotated set ahead of every other extension.

6 Discussion

The system result and the baseline result point in different directions and both are worth keeping. The agent outperforms single pass retrieval on matched questions, which supports the argument threaded through the ReAct line of work [7] that interleaving reasoning with tool calls lifts task accuracy. The retrieval hit rate falls within the range financial retrieval systems such as FinSage [3] report, so the hybrid design is competitive rather than merely functional.

The reranking result unsettles a common default. Cross encoder reranking is widely treated as a component that reliably helps, and on a large heterogeneous corpus it does. On a small uniform one it is a net cost to recall. Pradeep et al. [16] anticipate the mechanism in the abstract; this evaluation supplies a concrete instance and a design rule that follows from it.

The measurement finding deserves equal weight with the others, and it is the one most likely to generalise.

A faithfulness judge was truncating the retrieved context it saw to a fixed character cap, and that cap was smaller than the tool outputs produced for the Gulf Cooperation Council questions, which run long. The judge was therefore scoring faithfulness against a partial view of the evidence and understating it on precisely the corpus where the evidence was longest. Raising the cap corrected the measurement.

What matters is how the defect was found. No assertion in the evaluation harness caught it. It surfaced from reading what the judge was actually being passed. Where the metric is itself a model, validating the instrument belongs inside the experiment rather than before it, and a faithfulness judge that cannot see the whole evidence will penalise a grounded answer unfairly.

A boundary is worth drawing here. What this system offers is an audit trail over evidence, not insight into the generator. Which passage produced an answer, its document and section, its confidence, and the ranking steps that surfaced it are all visible. Why the model phrased its output as it did is not. Wilson [28] treats such coverage as partial, serving some stakeholders and not others, and that judgement is conceded here.

7 Limitations

The evaluation set is small. Thirty five Gulf Cooperation Council questions limit statistical power, as Section 5.5 makes explicit, and no amount of careful testing compensates for that.

A single annotator produced and verified the reference answers, leaving no basis for reporting agreement between raters. That review surfaced three incorrect entries. An instrument that yields three errors to one reviewer may well conceal others from the same reviewer, and a second independent pass would strengthen the ground truth.

Explainability here rests on design, not on evidence about readers. The attribution, the confidence value and the ranking steps ship with every answer, yet nobody in the intended audience was ever asked what those features let them establish. Building a system that displays its evidence and proving that the display changes a reader's ability to check it are separate claims, and only the first is supported.

The confidence value shown beside each passage is a retrieval confidence, not an answer confidence. It expresses how well a passage matches the query, not the probability that the answer built on it is correct, and a reader could reasonably misread it as the latter. Finally, the corpus is weighted toward large listed companies and may not transfer to small or unlisted firms, and FinanceBench questions cover 2016 to 2023 while the corpus covers 2024 to 2026, which is why that benchmark measures evidence overlap rather than answer accuracy here.

8 Future Work

Four directions follow, and one matters more than the rest. An explainability study with the intended readers would put source attribution, confidence, and the reasoning trace in front of board members and governance staff and measure whether those features change what a reader can verify. That is the evidence the explainability claim needs and this work does not supply. It is also the cheapest of the four to run, requiring participants rather than engineering.

The remaining three are ordinary increments: expanding the hand verified set so the paired comparison reaches conventional significance, extending the corpus toward small and unlisted issuers and additional Gulf Cooperation Council markets, and testing whether routed multi agent designs of the kind surveyed in Section 2.2 add value over a single agent with three tools.

9 Ethics Statement

No human participants took part in this study and no personal data is processed. Everything indexed is public corporate disclosure published by the issuers themselves through SEC EDGAR and Bahrain Bourse, so Bahrain Personal Data Protection Law, Law No. 30 of 2018, imposes no obligation that bites on this corpus. No formal ethics committee approval was required.

Two risks deserve naming. Automation bias is the first, and explainability can worsen it: a system displaying sources, scores, and a reasoning trace looks rigorous, and looking rigorous invites trust the underlying accuracy may not have earned. Fritz-Morgenthal et al. [27] and the CFA Institute report [28] both press the point that an explanation serves a reader only if that reader can evaluate it. The mitigations here are partial: the system abstains when its tools return nothing, prints that its output is not

financial advice, and shows the retrieved passage so a reader can consult the source rather than the summary.

The second concerns deployment rather than the study. A board member's queries reveal what that board is examining, and a production deployment sending those queries to a third party model provider would export a signal about the institution's own concerns. Nothing here resolves that. Generative AI was used during this project as a productivity aid for code implementation and for locating candidate sources, with every generated function tested against project data and every cited paper retrieved and verified against its original publication before citation.

10 Conclusion

Evaluating one agentic retrieval augmented system on two filing markets produced a result about evaluation itself. A closed book model answers 63.2 percent of United States questions and 11.4 percent of Gulf Cooperation Council questions, so single market evaluation measures a weaker problem than it appears to. The agent reaches 96.8 percent accuracy against 82.9 percent for single pass retrieval on matched items, with the direction of every discordant pair favouring the agent.

Two negative results are reported alongside those figures because they survived scrutiny: cross encoder reranking helps on a large heterogeneous corpus and costs recall on a small uniform one, and a faithfulness judge left unexamined misreported the metric it existed to measure. The most valuable single outcome of the project was not a feature but a defect, found by running a real query against a real company and noticing the answer was wrong.

Reproducibility Statement

Source code, the MIT licence, dependency versions, and every evaluation result file are released at the project repository, <https://github.com/hatemmsc-ops/CyberBoard-AI>, pinned at commit 9660fb81282a491d96f404a50422adfd8fb7eb9. Every figure in this paper is computed from a tracked result file rather than transcribed. Continuous integration runs the offline test suite on two Python versions and additionally asserts that the reported headline figures still match the result files they came from, so a reported number cannot drift from its evidence. Retrieval is deterministic; generation is served by a third party API and is not, which is why answer figures are reported as a five run mean with standard deviation.

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